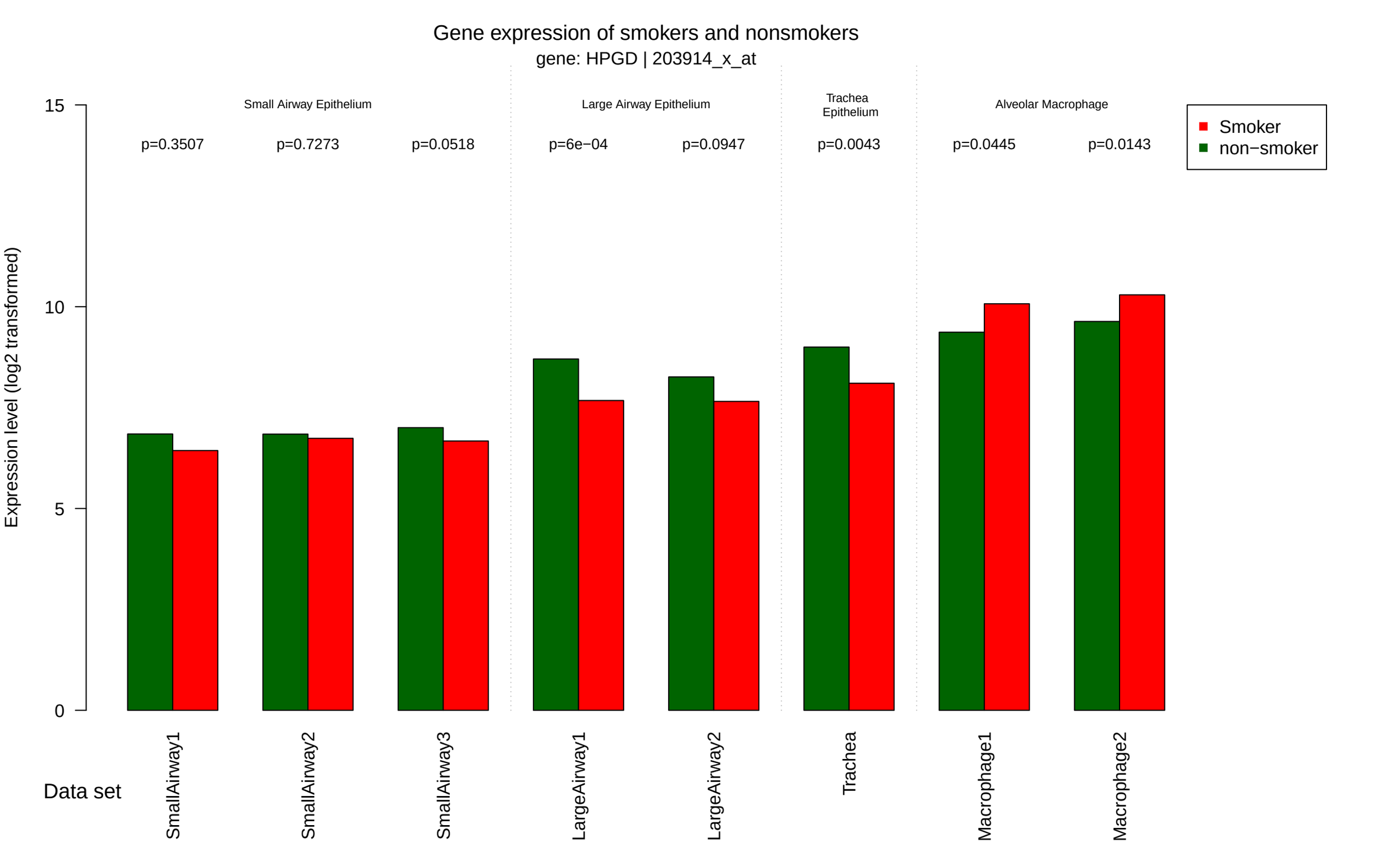
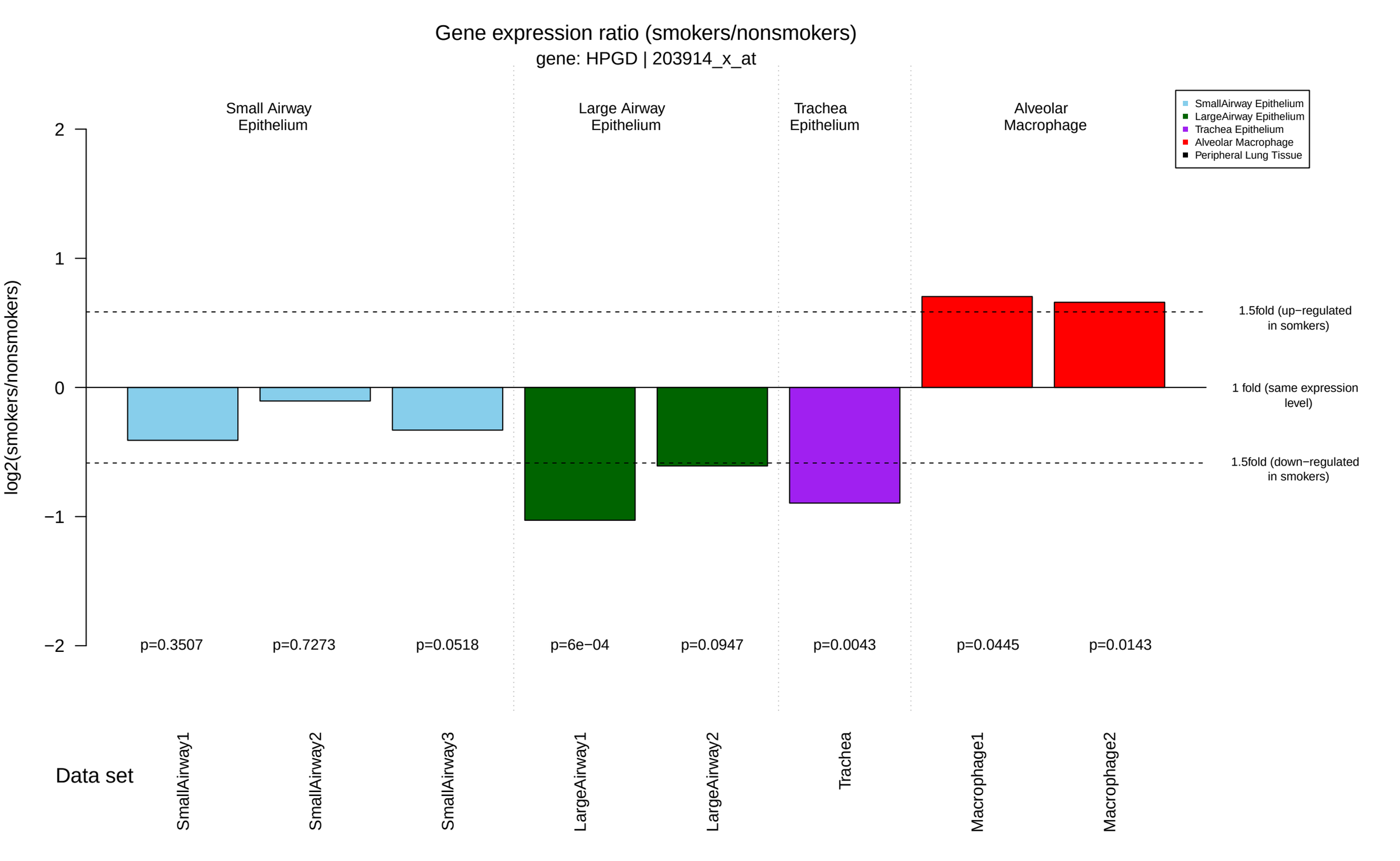
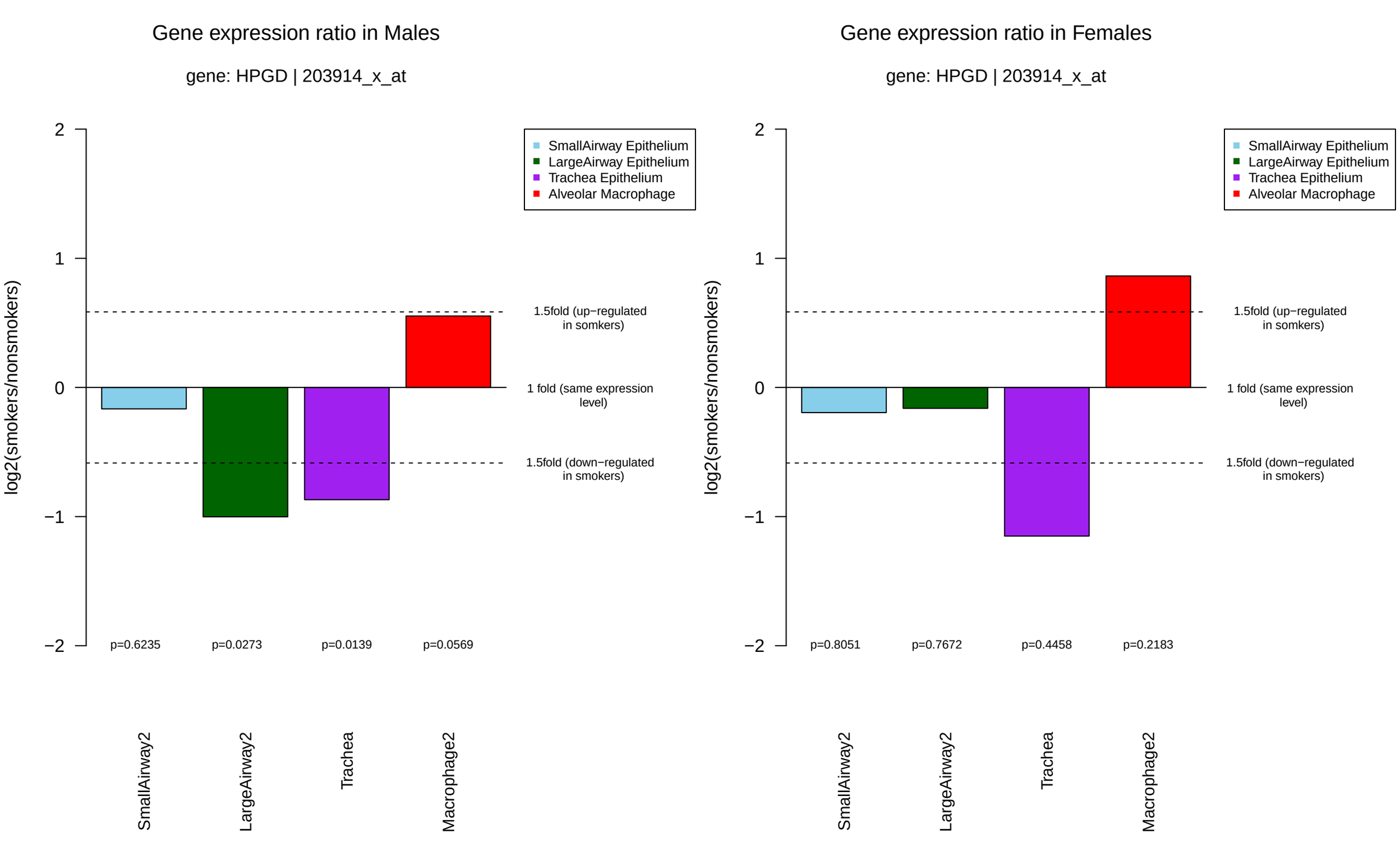
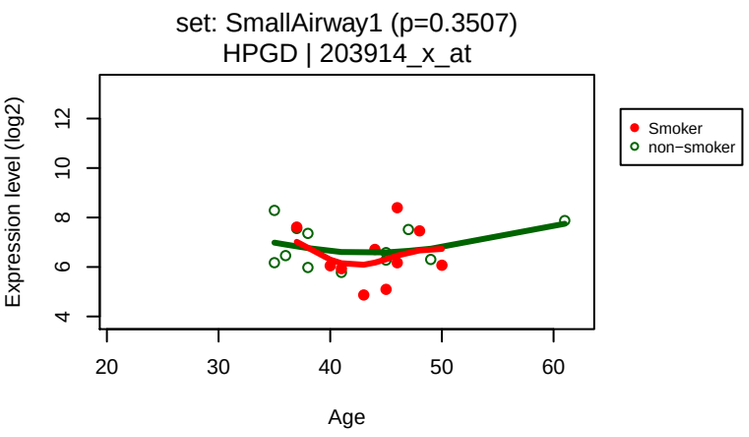
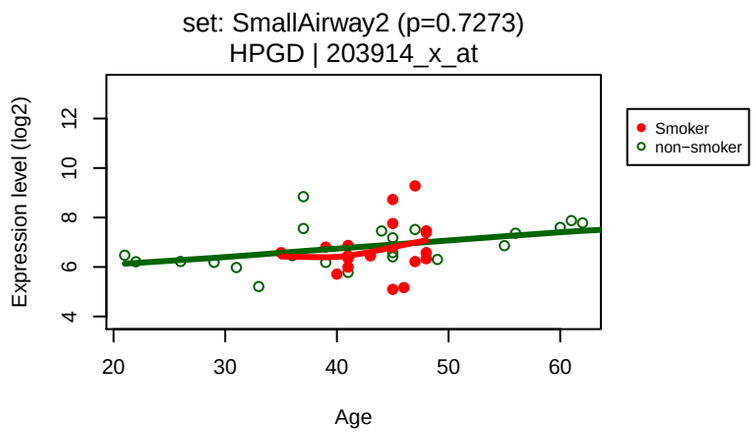
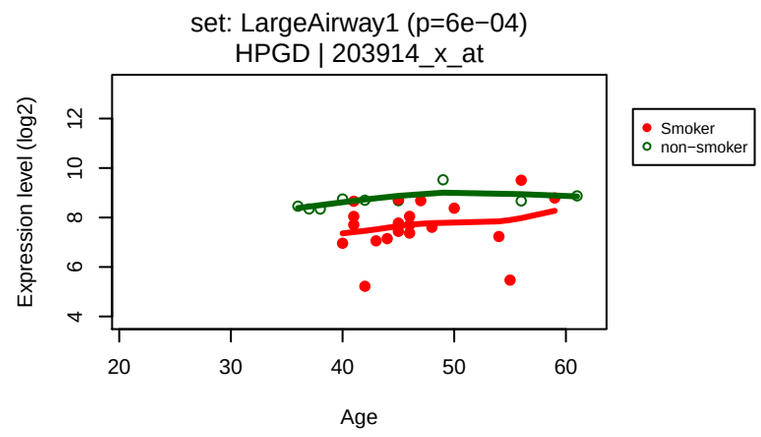

Gene expression of smokers and nonsmokers
gene: HPGD | 203914_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3507
p=0.7273
p=0.0518
p=6e−04
p=0.0947
p=0.0043
p=0.0445
p=0.0143
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

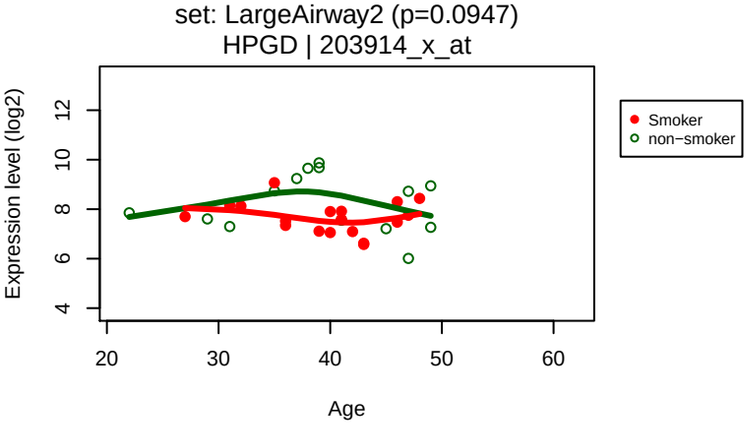

Gene expression ratio (smokers/nonsmokers)
gene: HPGD | 203914_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3507
p=0.7273
p=0.0518
p=6e−04
p=0.0947
p=0.0043
p=0.0445
p=0.0143
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

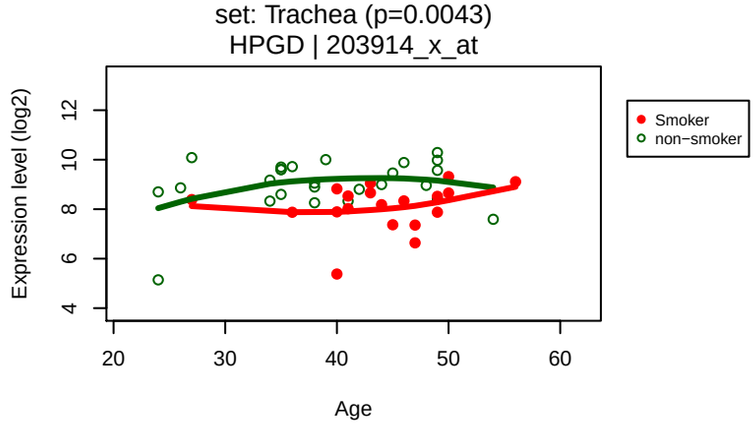

Gene expression ratio in Males
gene: HPGD | 203914_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6235
p=0.0273
p=0.0139
p=0.0569
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPGD | 203914_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8051
p=0.7672
p=0.4458
p=0.2183
SmallAirway2
LargeAirway2
Trachea
Macrophage2

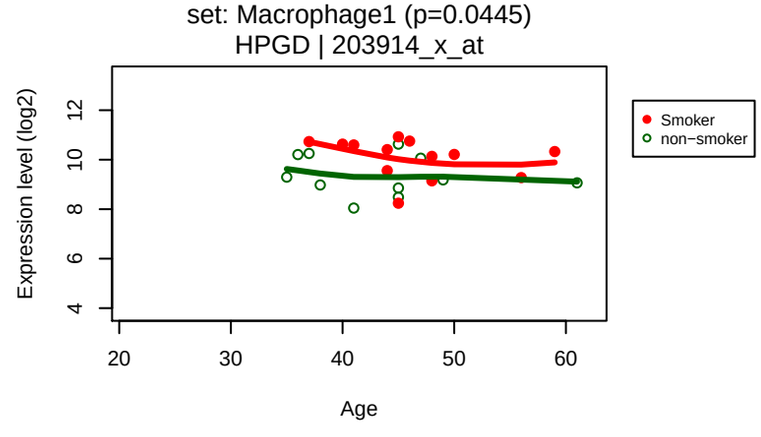

set: SmallAirway1 (p=0.3507)
HPGD | 203914_x_at
Expression level (log2)
Age
Smoker
non−smoker

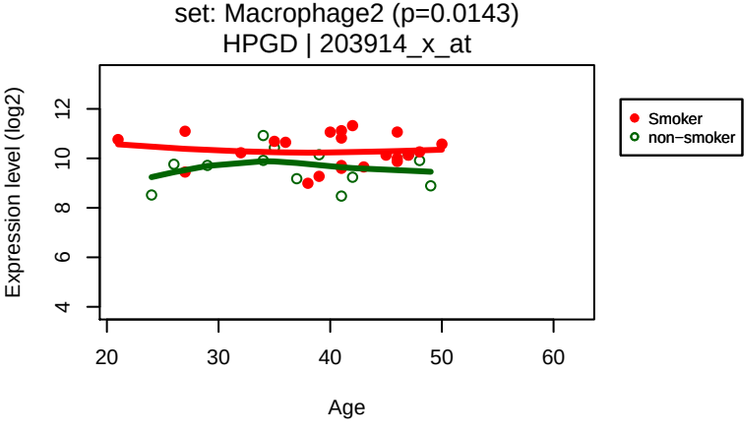

set: SmallAirway2 (p=0.7273)
HPGD | 203914_x_at
Expression level (log2)
Age
Smoker
non−smoker

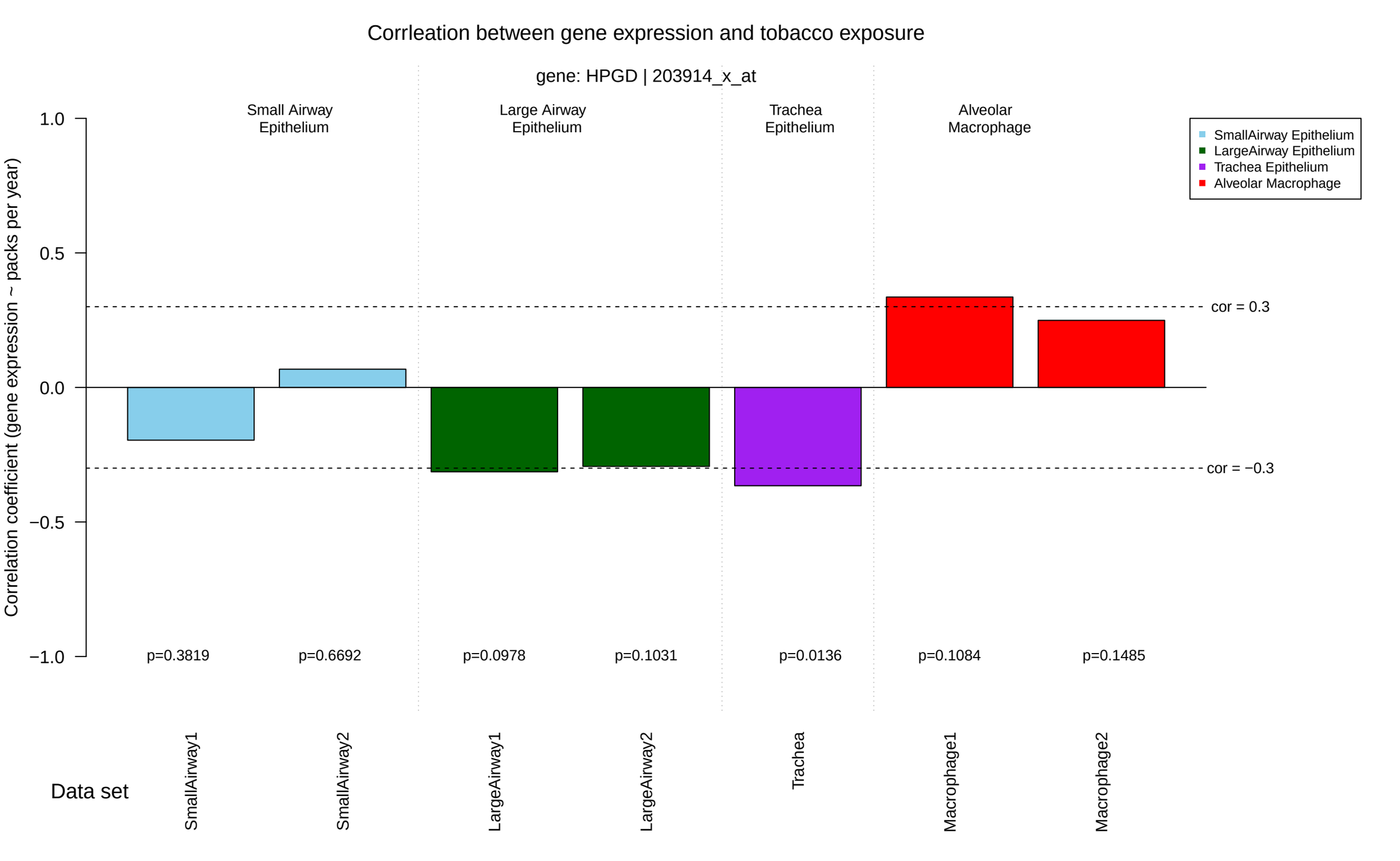

set: LargeAirway1 (p=6e−04)
HPGD | 203914_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0947)
HPGD | 203914_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0043)
HPGD | 203914_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0445)
HPGD | 203914_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0143)
HPGD | 203914_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPGD | 203914_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3819
p=0.6692
p=0.0978
p=0.1031
p=0.0136
p=0.1084
p=0.1485
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2